

GURVANTES, Mongolia

WMO: 443740

Lat: 43.2310N

Lon: 101.0430E

Elev: 5676

StdP: 11.92

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.4	-7.4	-28.3	1.4	3.1	-23.6	1.8	1.6	32.1	12.1	28.3	15.4	4.4	230	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	86.8	56.6	84.0	55.4	81.6	54.5	61.0	75.3	59.0	74.4	57.6	74.3	11.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.8	85.0	64.9	53.9	76.3	64.6	51.0	68.6	64.8	30.0	75.3	28.4	74.3	27.3	74.6	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
32.3	27.5	23.2	-15.4	90.7	4.4	2.7	-18.5	92.6	-21.1	94.2	-23.5	95.7	-26.7	97.7	
			-16.5	63.3	4.3	3.6	-19.6	65.9	-22.1	67.9	-24.5	69.9	-27.6	72.5	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.1	8.6	15.7	30.6	46.0	56.6	67.0	71.8	68.8	57.3	42.8	25.9	12.4	(6)
(7)		DBStd	23.42	8.83	10.39	10.32	9.48	8.31	6.21	5.35	5.59	7.18	8.71	8.53	8.40	(7)
(8)		HDD50	5219	1282	959	601	179	36	0	0	0	22	250	723	1167	(8)
(9)		HDD65	8860	1747	1379	1065	571	279	48	9	28	242	687	1173	1632	(9)
(10)		CDD50	2335	0	0	1	59	239	511	675	583	240	27	0	0	(10)
(11)		CDD65	501	0	0	0	1	18	108	219	145	10	0	0	0	(11)
(12)		CDH74	3777	0	0	0	4	130	854	1782	962	45	0	0	0	(12)
(13)		CDH80	894	0	0	0	0	14	192	497	190	1	0	0	0	(13)
(14)	Wind	WSAvg	9.9	7.0	8.8	11.3	12.6	13.0	10.6	9.6	9.0	9.9	9.9	8.0	(14)	
(15)	Precipitation	PrecAvg	3.8	0.0	0.1	0.2	0.2	0.3	0.6	1.0	0.5	0.2	0.1	0.1	(15)	
(16)		PrecMax	6.8	0.2	0.3	0.8	0.7	1.4	3.7	3.8	2.8	1.9	0.5	0.3	0.5	(16)
(17)		PrecMin	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.6	0.0	0.1	0.2	0.2	0.4	0.7	0.8	0.7	0.5	0.2	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	43.2	58.8	73.1	81.9	88.9	91.5	88.2	78.2	67.9	51.1	34.9	(19)
(20)		MCWB	25.9	30.1	39.3	47.2	50.5	56.0	59.2	58.2	50.8	45.4	36.2	25.3	(20)	
(21)		2%	DB	29.5	37.7	53.9	67.6	77.5	84.8	87.9	84.4	74.9	63.6	46.6	30.0	(21)
(22)		MCWB	22.1	26.8	36.0	44.0	49.3	54.2	57.5	56.4	49.8	43.4	33.4	22.3	(22)	
(23)		5%	DB	25.2	34.2	49.8	64.3	74.2	81.8	85.1	81.8	72.2	59.4	42.5	26.9	(23)
(24)		MCWB	19.0	24.7	33.8	42.4	47.9	53.1	56.3	54.9	48.7	41.3	31.0	20.5	(24)	
(25)		10%	DB	21.5	30.6	45.8	60.7	70.5	79.1	82.5	79.4	69.3	56.1	38.8	23.9	(25)
(26)		MCWB	16.5	22.6	31.7	40.4	46.2	52.2	55.6	54.5	47.6	39.4	28.9	18.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.4	30.8	39.9	48.4	52.2	58.9	64.4	64.5	54.6	46.6	36.9	25.3	(27)
(28)		MDCB	34.6	41.8	56.3	69.3	77.7	74.1	79.5	76.4	65.0	65.8	49.7	33.6	(28)	
(29)		2%	WB	22.4	27.4	36.7	44.6	50.1	57.0	61.8	61.1	52.7	44.3	33.8	22.7	(29)
(30)		MDCB	29.0	37.6	52.4	65.6	73.8	74.5	76.0	73.9	66.2	60.5	45.2	29.4	(30)	
(31)		5%	WB	19.3	24.9	34.3	42.7	48.5	55.6	59.7	58.6	51.0	42.3	31.4	20.6	(31)
(32)		MDCB	24.9	33.7	48.4	61.7	71.5	73.8	75.6	73.5	66.3	57.9	42.1	26.6	(32)	
(33)		10%	WB	16.6	22.7	32.1	41.0	47.1	54.4	58.3	57.1	49.5	40.1	29.1	18.6	(33)
(34)		MDCB	21.5	30.4	45.0	59.1	68.6	73.7	74.9	73.4	66.1	55.5	38.3	23.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	16.2	17.0	18.5	18.1	18.4	17.5	16.8	16.4	16.7	17.1	15.7	14.9	(35)	
(36)		5% DB	MCDDBR	18.5	19.7	20.6	20.2	20.3	19.4	19.0	18.6	18.6	19.0	18.5	16.9	(36)
(37)			MCWBR	13.3	13.1	12.0	10.8	9.7	7.7	6.7	6.8	7.6	10.0	10.9	12.2	(37)
(38)			MCDDBR	17.8	19.1	19.8	19.0	18.7	17.1	16.7	16.0	16.7	18.2	17.6	16.7	(38)
(39)		5% WB	MCWBR	13.1	12.9	11.8	10.7	9.5	7.0	6.8	7.2	7.7	10.0	10.7	12.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.265	0.303	0.345	0.387	0.402	0.418	0.409	0.390	0.356	0.317	0.283	0.262	(40)	
(41)		taud	2.350	2.253	2.132	2.067	2.081	2.113	2.192	2.228	2.278	2.323	2.359	2.351	(41)	
(42)		Ebn at Noon	279	282	281	278	276	271	272	274	275	273	268	268	(42)	
(43)		Edh at Noon	25	33	43	50	50	49	45	42	37	30	25	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	701	1039	1564	1988	2286	2269	2143	1964	1679	1270	830	630	(44)	
(45)		RadStd	66	91	94	88	103	140	106	76	63	35	51	55	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (2 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+101	(47)

Nomenclature: See separate page